

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Electrical Engineering) SEMESTER - 3 Summer 2026 (Supply.)

Course :Bachelor of Technology (Electrical Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 3

Subject Code & Name: 25AF1293PC303 - NETWORK THEORY

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

12

- 1 A voltage source having an open circuit voltage of 200 V and internal resistance of 50Ω is equivalent to a current source of _____
- a) 4A with 50Ω in parallel b) 4A with 50Ω in series
c) 0.5A with 50Ω in parallel d) none of the mentioned

Kirchhoff's voltage law is based on principle of conservation of _____

a) energy b) momentum c) mass d) charge

If source impedance is complex, then maximum power transfer occurs when the load impedance is _____ the source impedance.

- a) equal to b) negative of c) complex conjugate of
d) negative of complex conjugate of

- 4 Which of the following expression is true in case of open circuit parameters?

- a) $V_1 = Z_{11} V_1 + Z_{12} I_2$
b) $V_1 = Z_{11} I_1 + Z_{12} V_2$
c) $V_1 = Z_{11} I_1 + Z_{12} I_2$
d) $V_2 = Z_{11} I_1 + Z_{12} I_2$

If source impedance is complex, then maximum power transfer occurs when the load impedance is _____ the source impedance.

- a) equal to b) negative of
c) complex conjugate of d) negative of complex conjugate of

A network contains linear resistors and ideal voltage source s. If values of all the resistors are doubled, then voltage across each resistor is _____

- a) Halved b) Doubled
c) Increases by 2 times d) Remains same

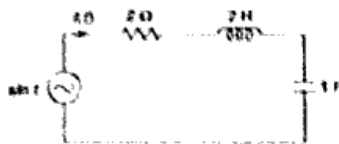
- 7 In the graph shown in Fig. one possible tree is formed by the branches 4, 5, 6, 7. Then one possible fundamental cutset is



Fig. 9.12.1

- (a) 1, 2, 3, 8 (b) 1, 7, 5, 6
(c) 1, 5, 6, 8 (d) 1, 2, 3, 7, 8

- 8 The number of independent loops for a network with n nodes and b branches is
(a) $n - 1$ (b) $b - n$
(c) $b - n + 1$ (d) independent of the number of nodes
- 9 The differential equation for the current $i(t)$ in the circuit of Fig. is



- (a) $2 \frac{d^2 i}{dt^2} + 2 \frac{di}{dt} + i(t) = \sin t$
(b) $\frac{d^2 i}{dt^2} + 2 \frac{di}{dt} + 2i(t) = \cos t$
(c) $2 \frac{d^2 i}{dt^2} + 2 \frac{di}{dt} + i(t) = \cos t$
(d) $\frac{d^2 i}{dt^2} + 2 \frac{di}{dt} + 2i(t) = \sin t$

- 10 Superposition theorem is not applicable to networks containing
(a) nonlinear elements (b) dependent voltage source
(c) dependent current source (d) transformers

11 Four resistances 80Ω , 50Ω , 25Ω , and R are connected in parallel. Current through 25Ω resistor is 4 A . Total current of the supply is 10 A . The value of R will be

- (a) 66.66Ω (b) 40.25Ω
(c) 36.36Ω (d) 76.56Ω

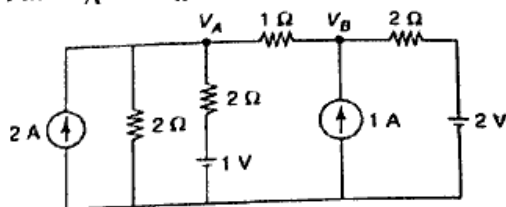
12 The nodal method of circuit analysis is based on

- (a) KVL and Ohm's law (b) KCL and Ohm's law
(c) KCL and KVL (d) KCL, KVL and Ohm's law

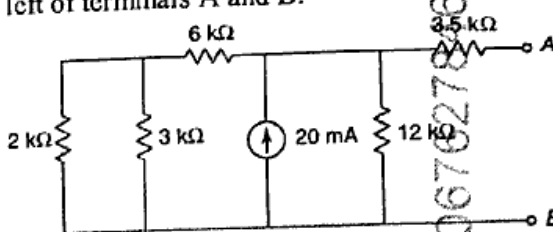
Q2 Solve the following

- A) Find V_A and V_B in the network shown in:

6



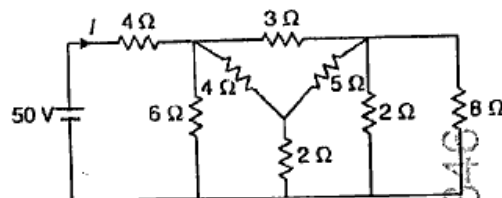
- B) Use source transformation to simplify the network until two elements remain to the left of terminals A and B.



- Q3. Solve the following.

- A) Determine the equivalent resistance in the given network:

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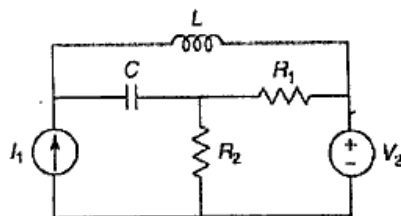
- B) Explain Maximum power transfer theorem and Nortans Theorem with its steps.

6

- Q4. Solve Any Two of the following.

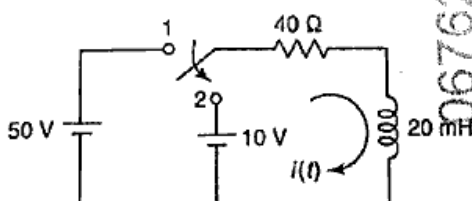
- A) For the networks shown in Fig. , write the incidence matrix, tieset matrix and f-cutset matrix.

6



- B) The network of Fig. is under steady state with switch at the position 1. At $t = 0$ switch is moved to position 2. Find $i(t)$. <https://www.batuonline.com>

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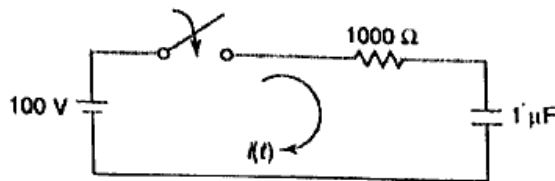


- C) Derive expression for transient response for series RL Circuit.

6

- Q5. Solve Any Two of the following.

- A) In the network of Fig. , the switch is closed at $t = 0$. With the capacitor uncharged, find value for i , $\frac{di}{dt}$ and $\frac{d^2i}{dt^2}$ at $t = 0^+$. 6

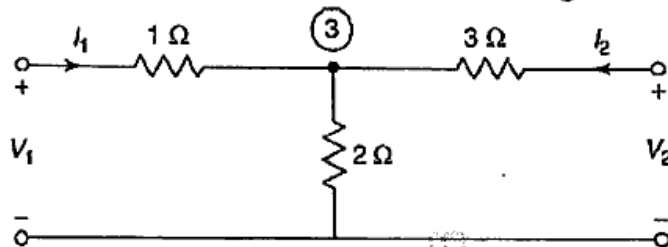


- B) A series RLC circuit has the following parameter values: $R = 10 \Omega$, $L = 0.01 \text{ H}$, $C = 100 \text{ mF}$. Compute the resonant frequency, bandwidth, and lower and upper frequencies of the bandwidth. 6

Describe Voltage, current and power relations in A balanced delta-connected load, also draw its phasor diagram for lagging power factor. 6

Q6. Solve Any Two of the following.

- A) Find Z-parameters for the network shown in Fig. 6



- B) Explain ABCD parameter for two port networks. 6
C) Explain resonant frequency, half power frequencies, bandwidth, quality factor for series RLC circuit. 6

*** End ***